

Naive Lie Theory 3.8 Exercises

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Notes

I used the result of 3.8.2 to prove 3.8.1, as the former is an equivalent definition of normal subgroup. There are many equivalent definitions of it, but this textbook introduces it insufficiently until this chapter, and thus I suggest one to check an algebra textbook. The most common definitions for H being normal are:

1. For any $h \in H$ and $g \in G$, $ghg^{-1} \in H$.
2. For any $h \in H$ and $g \in G$, there exists some $h' \in H$ such that $gh = h'g$.
3. For any $g \in G$, $gH = Hg$.
4. The cosets G/H is a group.

The original definition in 2.2 is 3 from the list above. I will use it to prove 1 and 2, which I used in 3.8.1. 3 says $\{hg|h \in H, g \in G\} = \{gh|h \in H, g \in G\}$, it's just equivalent to 2. For the equivalence between 1 and 2, 1 says $ghg^{-1} = h'$ for some $h' \in H$, multiplying g on the right, which is again directly equivalent to 2.

3.8.1

For the sake of contradiction, say $G/Z(G)$ has a nontrivial normal subgroup $N = \{hZ(G)|h \in H, H \subseteq G\}$, where $H \neq \emptyset$ and $H \neq \{e\}$, which means for any $g \in G$, $h \in H$, $ghg^{-1}Z(G) \in HZ(G)$. H must be a subgroup of G for N to be a subgroup of $G/Z(G)$. Writing $Z(G)$ as a representative, for any $z \in Z(G)$, $ghg^{-1}z = ghzg^{-1} = h'z'$. Consider the set $\{hz = zh|h \in H, z \in Z(G)\}$. It's a normal subgroup of G . Contradict.

3.8.2

See Notes.

3.8.3

The identity component is the set of matrices A such that there exists a continuous path from I to A . It's a subgroup. Check 3.2 exercise for the proof.

Let G° be the identity component. Obviously $I \in G^\circ$, and for any $A \in G$ $AIA^{-1} = I$. For any $h \in G^\circ$ with a path $\gamma : [0, 1] \rightarrow G$, where $\gamma(0) = I$ and $\gamma(1) = h$, conjugate the whole path so that $\bar{\gamma}(t) = g\gamma(t)g^{-1}$, where $\bar{\gamma}(0) = I$ and $\bar{\gamma}(1) = ghg^{-1}$, thus $ghg^{-1} \in G^\circ$.

3.8.4

Not an interesting problem. I will use the fact that $\mathbb{Q}$ is dense in $\mathbb{R}$, but proving it requires much irrelevant efforts. Consider the trivial continuous surjections $\mathbb{Q} \rightarrow H$ and $\mathbb{R} \rightarrow S^1$. Identify an arc as the image of some open interval (a, b) . By density of $\mathbb{Q}$ in $\mathbb{R}$, there exists $q \in \mathbb{Q}$ such that $a < q < b$. Thus the image of q lies in the arc. For disconnectedness, take any two distinct elements of H , say their preimages are q_1, q_2 . Another standard fact is that there exists $r \in \mathbb{R} - \mathbb{Q}$ such that $q_1 < r < q_2$. Lift it up to show disconnectedness.

3.8.5

As shown in the textbook in this chapter, given $SO(2)$ is path-connected, a normal and totally disconnected subgroup is in the center. $Z(SO(n)) \in \{\{I\}, \{\pm I\}\}$, neither of each can be dense, let alone a subset of each.